

Quantum Spirals and Multi-Reality Space-Time: The Emergence of Infinity in the Expanding Void

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Abstract

This paper explores the emergence of infinity within quantum mechanics through the interplay of *two infinite entanglements* embedded in an expanding void. By redefining “nothingness” as the source of infinite potentialities and the “void” as the dynamical substrate of unobserved expanding space-time, we propose a framework where quantum phenomena emerge dynamically. We develop an extended Schrödinger equation incorporating these concepts and clarify the assumptions behind the entanglement interaction terms. Additionally, we connect these ideas to cosmological inflation, interpretations of quantum measurement, and observer-consciousness interplay. Our work bridges mathematical physics, metaphysics, and philosophy, outlining possible avenues for empirical validation.

1 Introduction

Quantum mechanics challenges classical reality with phenomena such as entanglement, superposition, and wave function collapse. While interpretations like Copenhagen [1] and Many-Worlds [2] provide explanatory frameworks, the underlying origin of quantum behavior and its connection to cosmology and consciousness remains open.

Here, we propose a novel conceptual and mathematical framework founded on two main ideas:

1. The **duality of nothingness and void**: nothingness represents the infinite set of potential quantum states (unmanifested possibilities), while the void corresponds to a physically expanding, unobserved space-time substrate.
2. The **interaction of two infinite entanglements**, represented as dynamically coupled quantum states evolving within the expanding void, forming complex structures we term *quantum spirals*.

This framework aims to unify quantum foundations, cosmological expansion, and metaphysical considerations of consciousness and observation.

2 The Duality of Nothingness and Void

2.1 Physical Interpretation

Nothingness is conceptualized as the totality of all possible quantum states prior to actualization — mathematically an infinite-dimensional Hilbert space of potential wave functions.

Void is modeled as an expanding spatial background, akin to cosmological inflation [3], providing a time-dependent metric that affects quantum state evolution. The void represents unobserved space-time regions that grow exponentially, modeled by a potential $V(t) = V_0 e^{\lambda t}$.

Together, they describe a system where infinite possibilities exist (nothingness), and the physical substrate (void) evolves to manifest those possibilities into observable quantum phenomena.

2.2 Philosophical Context

Philosophically, this resonates with metaphysical traditions such as Advaita Vedanta's *Brahman* and Buddhist *Shunyata* (emptiness) [5], interpreted here as infinite potential (nothingness) realized through dynamic space-time (void).

3 Mathematical Framework

We extend the conventional Schrödinger equation to capture the interaction of two entangled quantum states ψ_1, ψ_2 within an expanding void:

$$i\hbar\frac{\partial}{\partial t}\psi = \hat{H}\psi + \alpha\nabla^2\psi + \beta f(\psi_1, \psi_2), \quad (1)$$

where

$$f(\psi_1, \psi_2) = |\psi_1|^2\nabla\psi_2 - |\psi_2|^2\nabla\psi_1.$$

Physical system: This model describes a pair of coupled quantum fields or particles interacting through their probability amplitudes’ gradients, influenced by spatial curvature induced by the expanding void.

Assumptions:

- The void expansion affects quantum evolution via the spatial Laplacian term weighted by α .
- The nonlinear coupling term $f(\psi_1, \psi_2)$ models dynamical entanglement feedback between the two states, capturing information exchange beyond linear superposition.
- The potential $V(t) = V_0 e^{\lambda t}$ encodes exponential expansion, motivated by cosmological inflationary models [3, 4].

3.1 Example Simulation

Consider two Gaussian wave packets ψ_1, ψ_2 initially localized but interacting via f . Numerical integration of (1) shows spiral-like interference patterns whose scale grows with time due to the exponential potential $V(t)$. These *quantum spirals* represent evolving pathways of entanglement in the expanding void.

4 Terminology Precision

Terms like “two infinite entanglements” and “quantum spirals” serve as conceptual metaphors but are grounded in mathematical definitions above. *Two infinite entanglements* refers to two continuously interacting quantum states with infinite degrees of freedom evolving nonlinearly. *Quantum spirals* are emergent phase-space trajectories arising from these interactions under the expanding background.

5 Observer, Measurement, and Wave Function Collapse

Wave function collapse is modeled as a projection operator $\hat{P}$ selecting a realized reality from the infinite superposition:

$$\Phi_{\text{collapsed}}(x, t) = \hat{P}\Phi(x, t),$$

where $\Phi = \psi_1 + \psi_2$.

Here, the observer's consciousness acts as the active selector from the infinite possibilities (nothingness) within the expanding void, stabilizing a particular quantum spiral. This aligns with interpretations emphasizing observer-participation in reality formation [6].

6 References and Related Work

Beyond foundational works [1, 2], this framework builds on recent developments in:

- Quantum decoherence and observer effects [7]
- Quantum gravity and cosmological inflation [3, 4]
- Consciousness and quantum foundations [8, 9]

These connections strengthen the physical plausibility and philosophical grounding of our model.

7 Testable Predictions and Future Directions

The model suggests observable consequences:

- Modified interference patterns in quantum systems embedded in expanding or curved backgrounds.
- Correlations between entangled particles reflecting nonlinear gradient couplings.
- Possible signatures of quantum spiral dynamics in cold atom experiments or quantum simulators.

Future work will focus on detailed numerical simulations, experimental proposals, and integrating quantum gravity insights.

8 Conclusion

This work proposes a mathematically grounded and philosophically enriched framework linking infinite entanglements, expanding void, and observer consciousness to explain quantum phenomena and cosmic emergence. By clarifying assumptions and defining novel concepts precisely, we open pathways for empirical and theoretical advances bridging physics, metaphysics, and consciousness studies.

References

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